

Air Quality & Microclimates in Prague

Data Processing in Python - Work-in-Progress Consultation

1 Project Goal

This project looks at how local weather conditions (temperature, humidity, pressure, rain, and wind) affect air pollution (PM_{10} and NO_2) in different parts of Prague. The end goal is to build an interactive online dashboard showing the interplay between pollution and weather.

2 What We Have Done So Far

- **Automated Data Downloads:** We wrote Python scripts that automatically download raw data from the Czech Hydrometeorological Institute for both air quality sensors and weather stations.
- **Filtered and Processed Data:** We filtered only for Prague stations and explored the characteristics of data at hand - we had to drop two air quality stations due to missing data.
- **Matched Mismatched Stations:** Air quality sensors and weather sensors are not placed at the exact same spots. We calculated the physical distances between all stations and used a "Nearest Neighbor" approach to match every air quality station to its closest weather counterpart.
- **Cleaned the Datasets:** We combined the datasets onto a single timeline, fixed missing values (nulls/negatives), and set up a clean, local file structure to work from.

3 Current Steps & Model Setup

- **Creating explanatory variables:** We are converting wind speed and direction angles into vectors, aggregating precipitation, adding lags, changes and averages to get more characteristics.
- **Converting Time Variable:** In addition, we would turn hours and months into cyclical waves (sine/cosine) so the computer understands that 23:00 (11 PM) is close to 00:00 (Midnight) to make the relationship comparable over the day.
- **Predictive Machine Learning:** We have drafted a **Random Forest Regressor** model. So far, we tried basic specification and time series split validation.

4 Consultation Questions

Question 1: Distant Weather Stations

- Our station-matching script paired everything up, but a few isolated air quality stations are quite far away from any weather station.
- If we keep them, the weather data might not be accurate for that specific street. If we drop them, we lose coverage of certain Prague districts.
- Should we enforce a strict rule (e.g., maximum distance of 5 km) and drop the far ones, or should we use an average of the 3 nearest stations to estimate the weather at that location?

Question 2: Data Frequency (Hourly vs. Daily)

- For the modelling to capture dynamics of sudden wind gusts or heavy rain bursts, hourly data is perfect. But for displaying a clean map on our dashboard, daily averages would look much better.
- We plan to use a dual approach: keep hourly data for training the model, but compress it into daily averages for the dashboards. Does this sound reasonable?

Question 3: Time Period Extension

- Right now, our code runs smoothly on a 1-month pilot slice of 2025 data.
- We plan to expand this download to cover a longer time frame, e.g. 1.5 year, so our model can accurately identify seasonal patterns. Is that a good benchmark for a project of this scale? There are no data directly available before 2025 for emissions.

5 Next Steps

1. **Expand the dataset** to the full 1.5 year window based on the feedback.
2. Set up **more models** to discover the interplay between variables (hard to do it causally though).
3. Build the **interactive dashboard** in Streamlit.